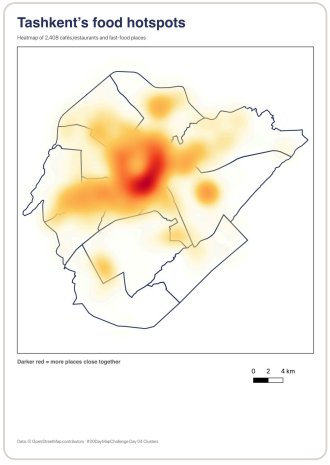
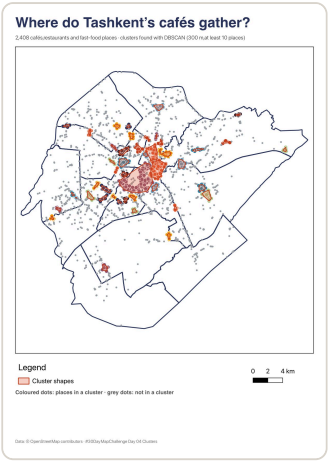
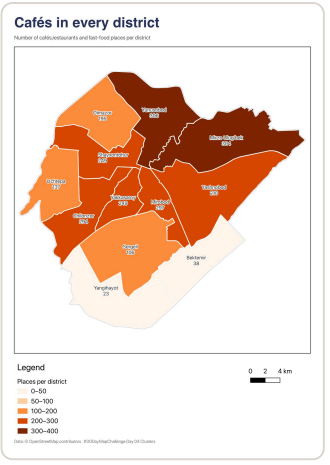
🫧 Day 04 · Clusters: where do Tashkent's cafés gather?

A beginner tutorial with QGIS, the free desktop map program used by professionals. No coding, no experience needed.

You will put **2,408 cafés, restaurants and fast-food places** from OpenStreetMap on a map of Tashkent, see where they crowd together with a **heatmap**, let QGIS find the **clusters** for you with the **DBSCAN** tool, draw a shape around each cluster, count the places in every **district**, and finish with a printable map.

⏱️ About 45 minutes (plus 10 minutes to install QGIS) · 💻 Windows, macOS or Linux · 🔑 No account needed

Heatmap	DBSCAN clusters	Places per district
		

📚 Part 0 · The ideas you need

Points. Each café is one point with a location and a few columns (`name` , `kind`).

Cluster. A group of points that are much closer to each other than to the rest. Clusters show where life happens: shopping streets, the city centre, a market.

Heatmap. Every point spreads a little "heat" around it. Where many points are close, the heat adds up and turns red. It's a quick way to *see* clusters.

DBSCAN. A tool that *finds* clusters with two simple rules:

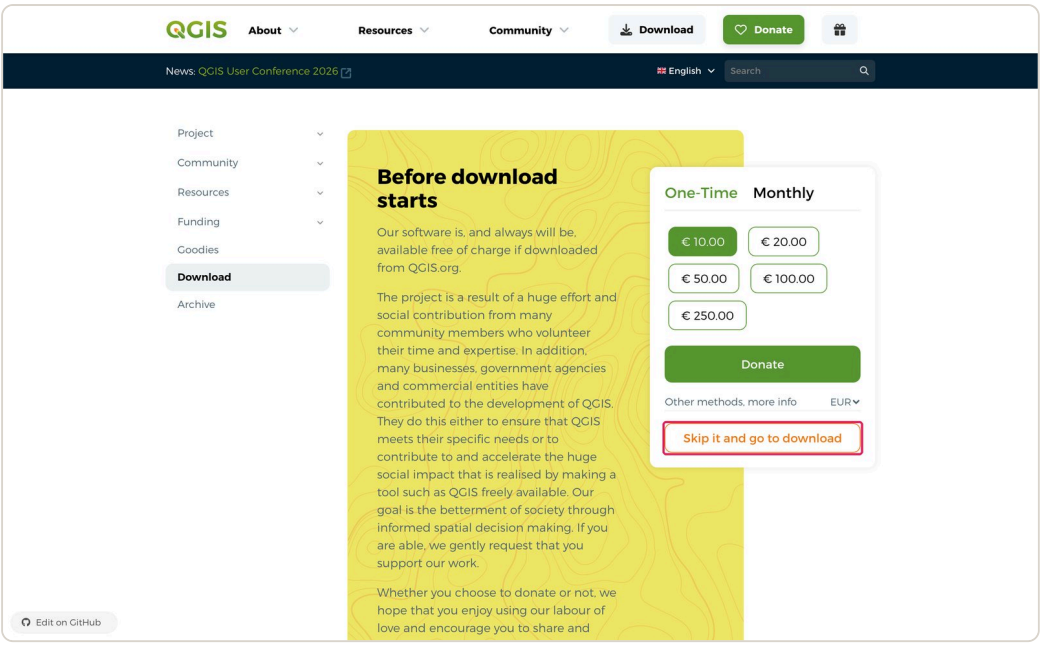
1. **Distance:** two places are neighbours if they are less than, for example, **300 m** apart.
2. **Minimum size:** a group needs at least, for example, **10 places** to count as a cluster.

Places that are too far from any group are **noise**: they don't belong to a cluster.

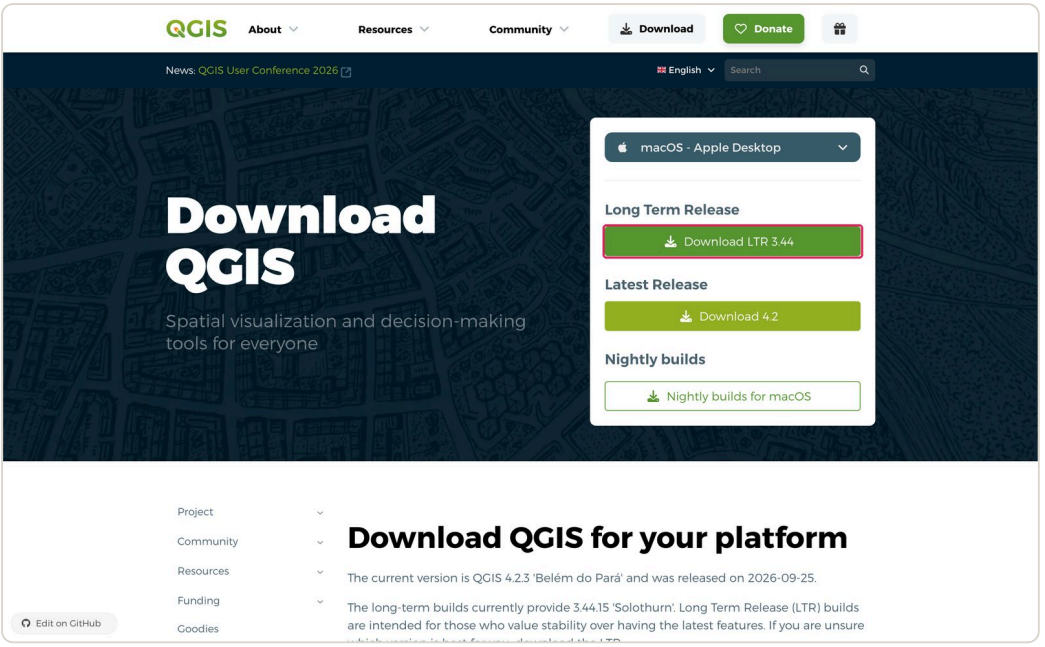
Projection. Our data uses degrees (longitude and latitude). DBSCAN needs **metres**, so we reproject the points into a metre-based system first.

💾 Part 1 · Install QGIS

1. Go to **qgis.org/download**. The page first asks for a donation. QGIS is free: click **Skip it and go to download**.



2. The page picks your system (Windows or macOS) automatically. Click **Download LTR** (Long Term Release, the most stable version).



3. Install it like any other program: on **Windows** run the `.msi` file and click *Next* until it finishes. On **macOS** open the `.dmg` file and drag QGIS into *Applications*.

> 💡 The screenshots in this tutorial come from QGIS 3.34 on a Mac. Newer versions (3.44 and later) look almost the same. On Windows the menus (**Project**, **View**, **Layer**, **Vector**…) are at the top of the QGIS window, and on a Mac they are at the top of the screen.

📥 Part 2 · Get the data

Download both files into one folder:

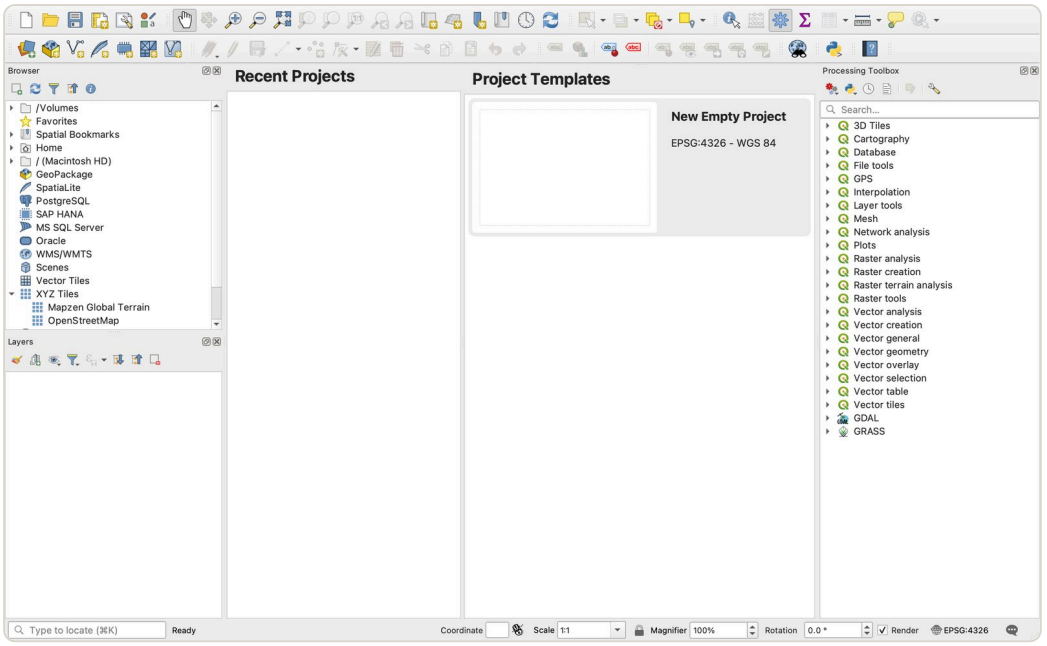
File	What's inside
`tashkent_food_places.geojson`	2,408 cafés, restaurants and fast-food places (columns `name`, `kind`)
`tashkent_districts.geojson`	the 12 districts of Tashkent City (column `district`)

⬇️ Or both at once: **day04_clusters_data.zip**, then unzip it.

To download one file from GitHub: click its name, then the **Download raw file** button (⬇) at the top right.

🖥️ Part 3 · Meet QGIS

Open QGIS. You see the start screen:

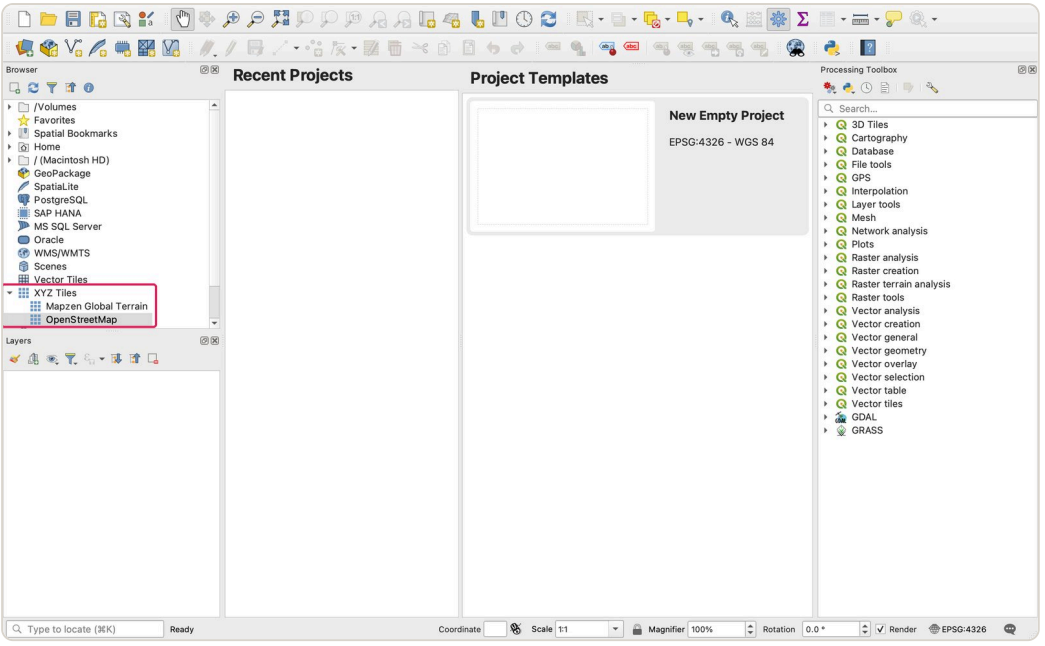


The four parts you will use:

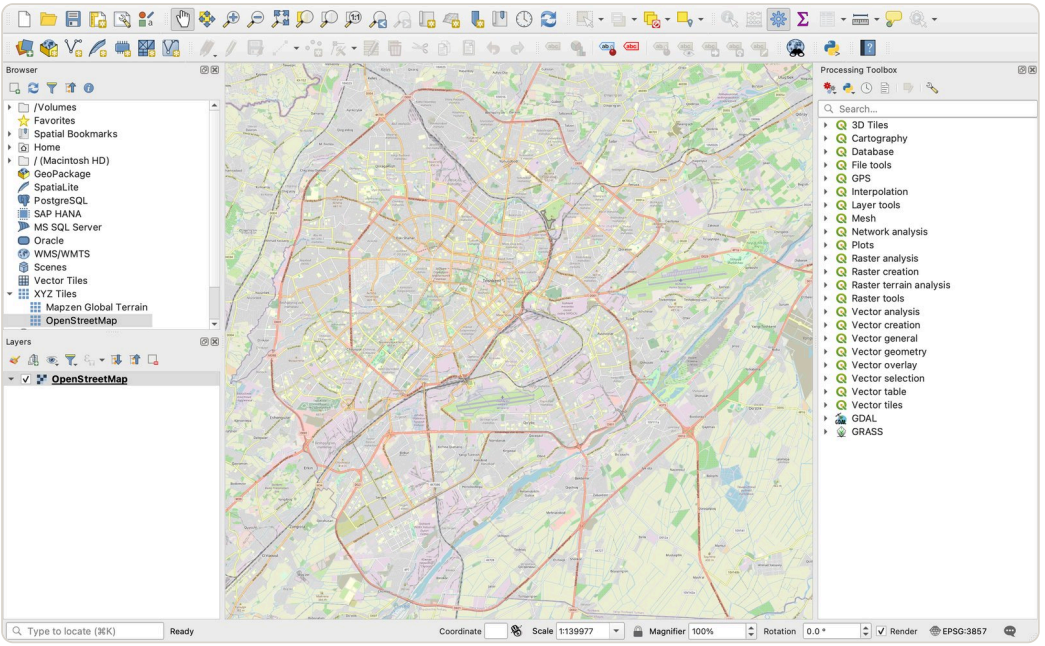
- **Browser** (top left): your files and online maps.
- **Layers** (bottom left): everything on your map. Layers at the top are drawn on top.
- **Map** (centre): the map itself. Scroll to zoom, hold the mouse button and drag to move.
- **Processing Toolbox** (right): hundreds of tools with a search box. If you don't see it, open **Processing → Toolbox**.

🗺️ Part 4 · Add a background map

In the **Browser**, click the small arrow next to **XYZ Tiles**, then **double-click OpenStreetMap**.

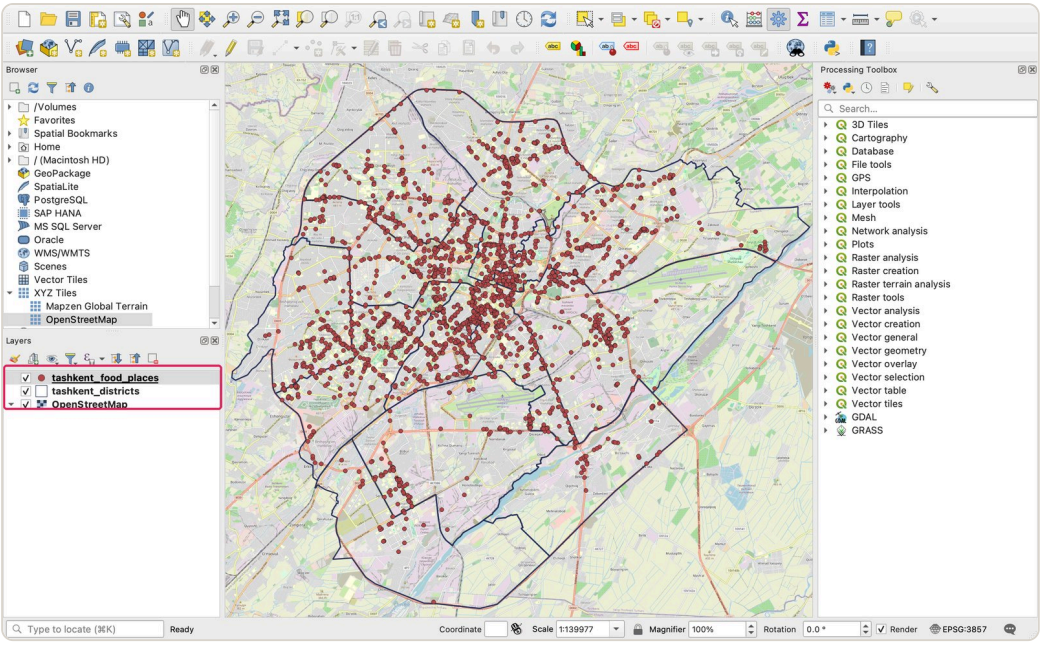


The world map appears. Zoom in to Tashkent with the mouse wheel, or wait for Part 5: once the districts are loaded, right-click `tashkent_districts` → **Zoom to Layer** jumps straight there.

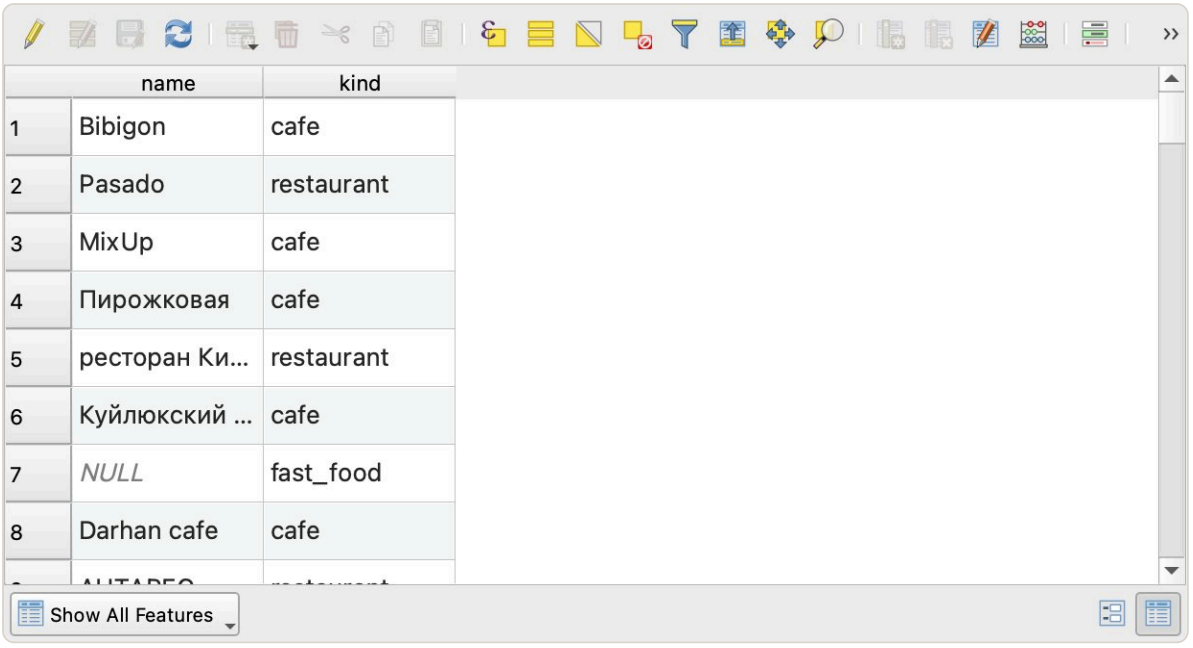


📍 Part 5 · Add the cafés and districts

1. Drag `tashkent_districts.geojson` and then `tashkent_food_places.geojson` from your folder into the QGIS map. They appear in the **Layers** panel.



2. Look inside: right-click `tashkent_food_places` → **Open Attribute Table**. Every row is one place, with its `name` and `kind` (`cafe`, `restaurant` or `fast_food`). Some names are empty (`NULL`) because nobody added them to OpenStreetMap yet.



	name	kind
1	Bibigon	cafe
2	Pasado	restaurant
3	MixUp	cafe
4	Пирожковая	cafe
5	ресторан Ки...	restaurant
6	Куйлюкский ...	cafe
7	*NULL*	fast_food
8	Darhan cafe	cafe

Show All Features

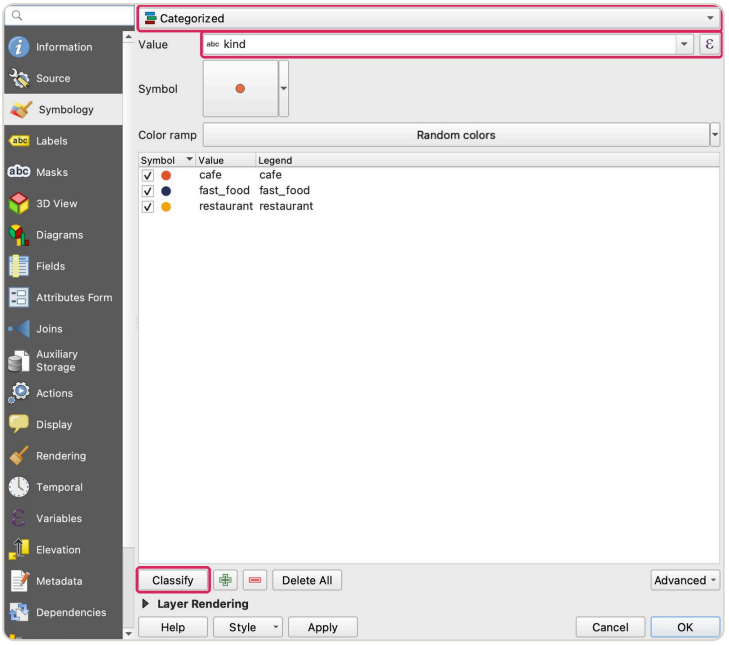
3. Close the table.

> 💡 If the districts cover the map in one colour: double-click `tashkent_districts` → **Symbology** → click **Simple Fill** → set **Fill style** to *No Brush*. Now only the borders are drawn.

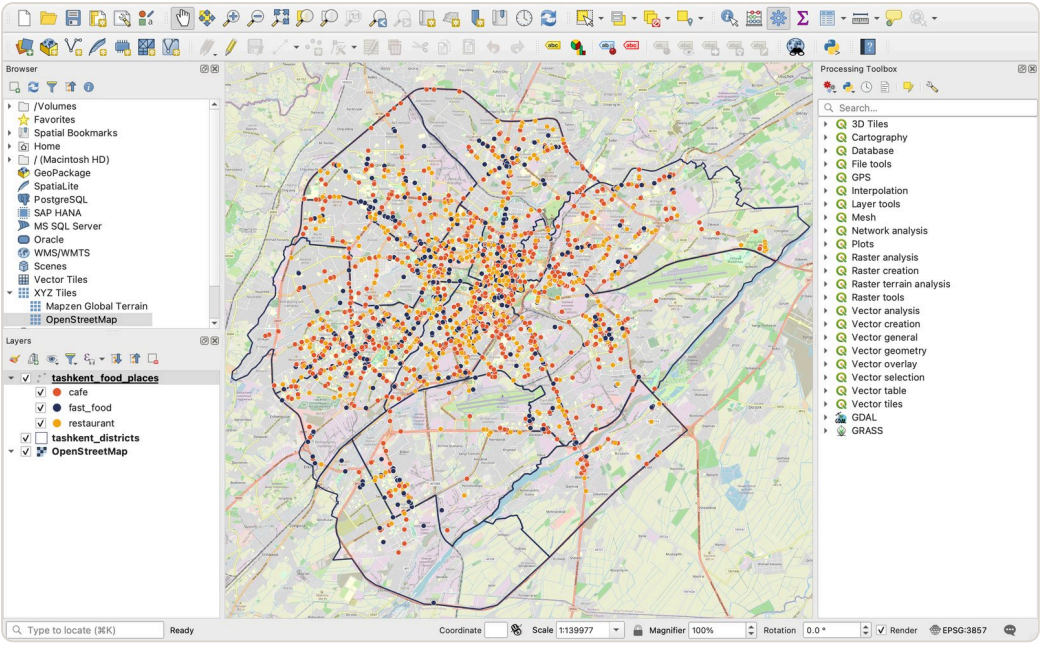
🎨 Part 6 · Colour the places by kind

1. **Double-click** `tashkent_food_places` in the Layers panel. The **Layer Properties** window opens.
2. Click **Symbology** on the left.

3. At the top, change **Single Symbol** to **Categorized**.
4. Set **Value** to `kind`.
5. Click **Classify**, then **OK**.

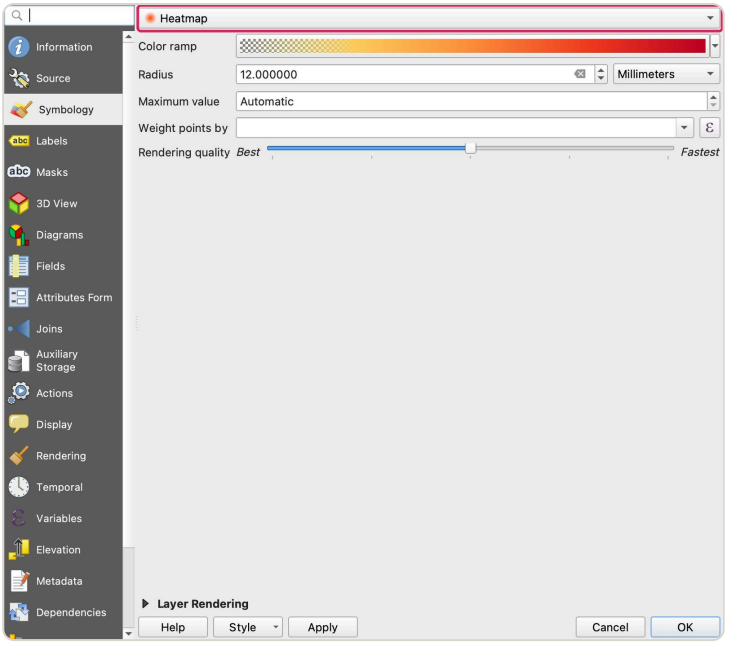


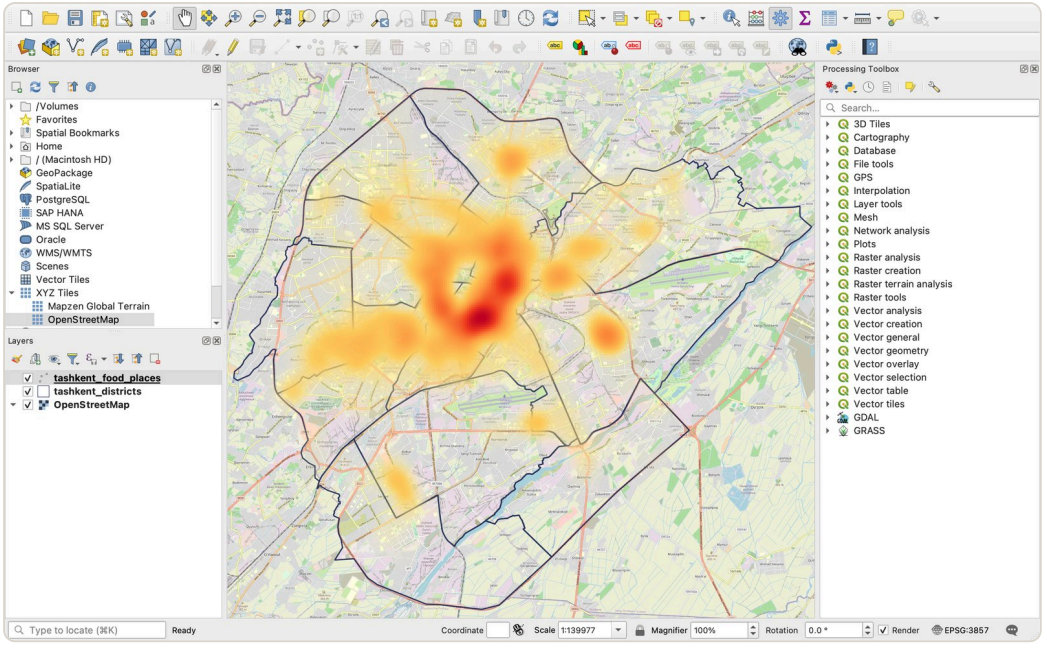
Cafés, restaurants and fast food now have their own colours:



🔥 Part 7 · A heatmap: see the hotspots

1. Double-click `tashkent_food_places` again → **Symbology**.
2. Change **Categorized** to **Heatmap**.
3. Pick a **Color ramp** you like (click the small arrow next to it, e.g. *Reds* or *Magma*) and set **Radius** to about `12` millimetres. Click **OK**.





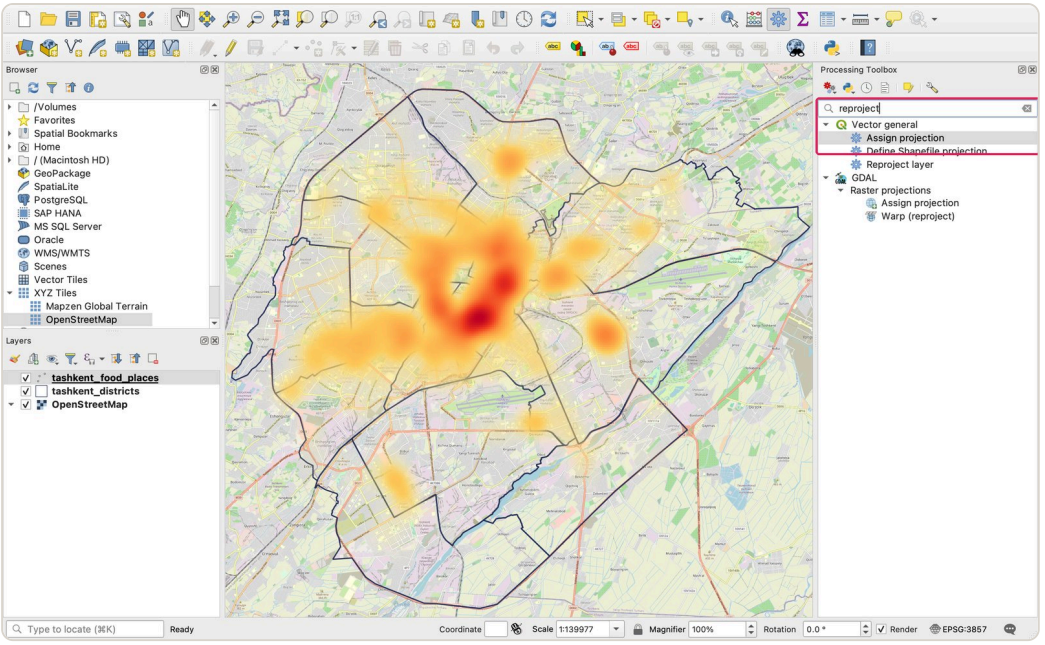
The hottest spot is the **city centre**. Smaller hotspots appear along big streets and in the neighbourhoods.

> 💡 To make the background quieter: double-click `OpenStreetMap` → **Transparency** → set **Global Opacity** to about 50%.

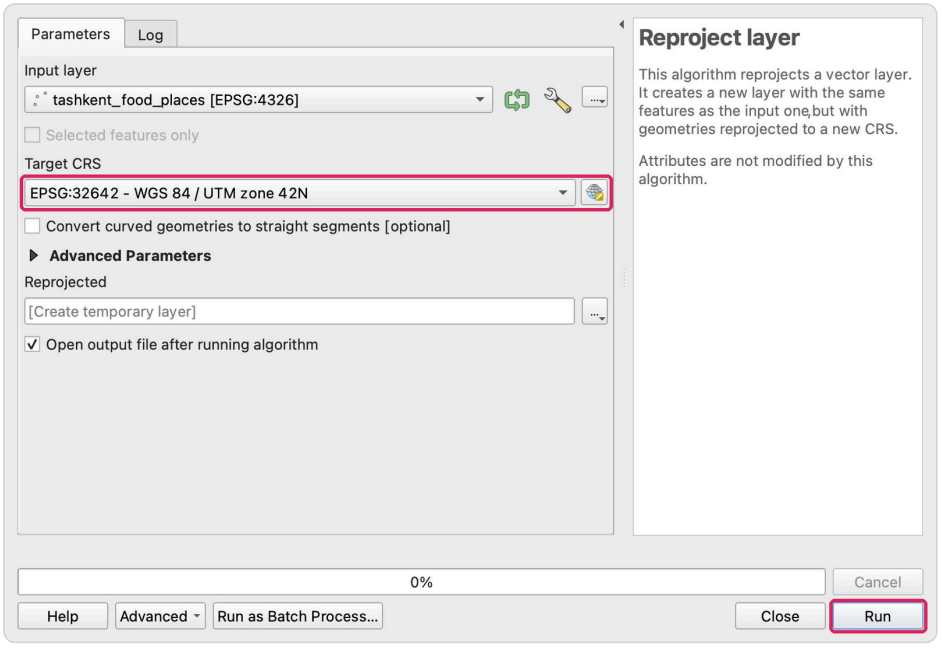
🔪 Part 8 · From degrees to metres (reproject)

DBSCAN measures distance in the units of the layer. Our points are in **degrees**, so "300" would mean 300 degrees, almost the whole Earth! Let's convert them to **metres**.

1. In the **Processing Toolbox**, type `reproject` in the search box and double-click **Reproject layer**.

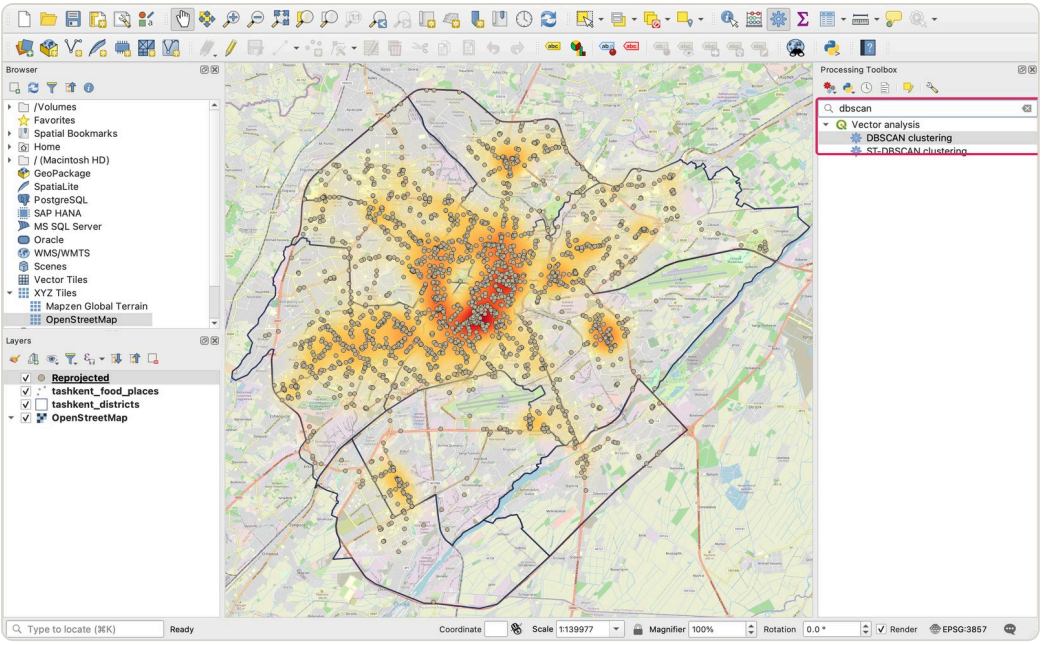


2. **Input layer:** `tashkent_food_places`.
3. **Target CRS:** click the small globe button and search for `32642`. Choose **WGS 84 / UTM zone 42N** (EPSG:32642), the metre-based system for Tashkent.
4. Click **Run**, then **Close**. A new layer called **Reprojected** appears.

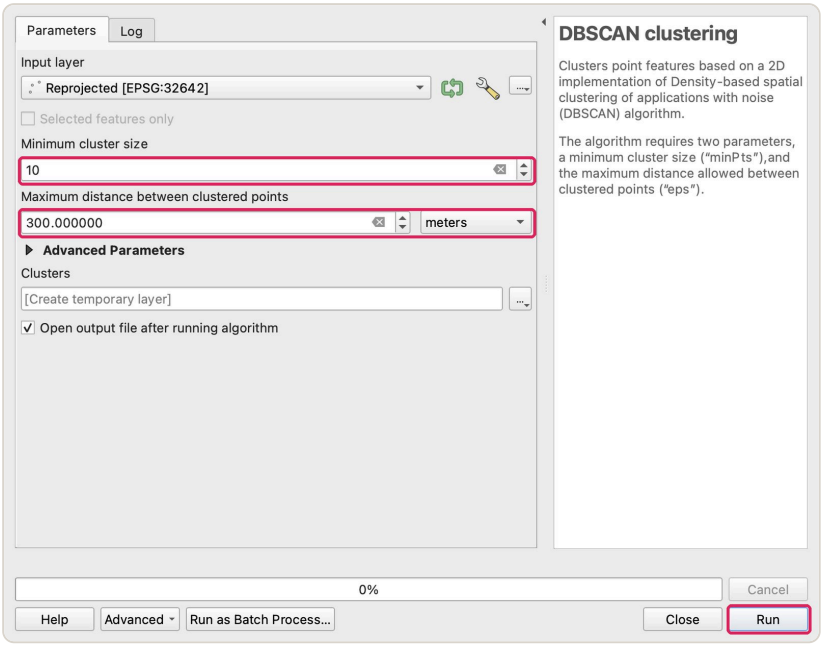


Part 9 · Find the clusters (DBSCAN)

1. In the Processing Toolbox, search `dbscan` and double-click **DBSCAN clustering**.

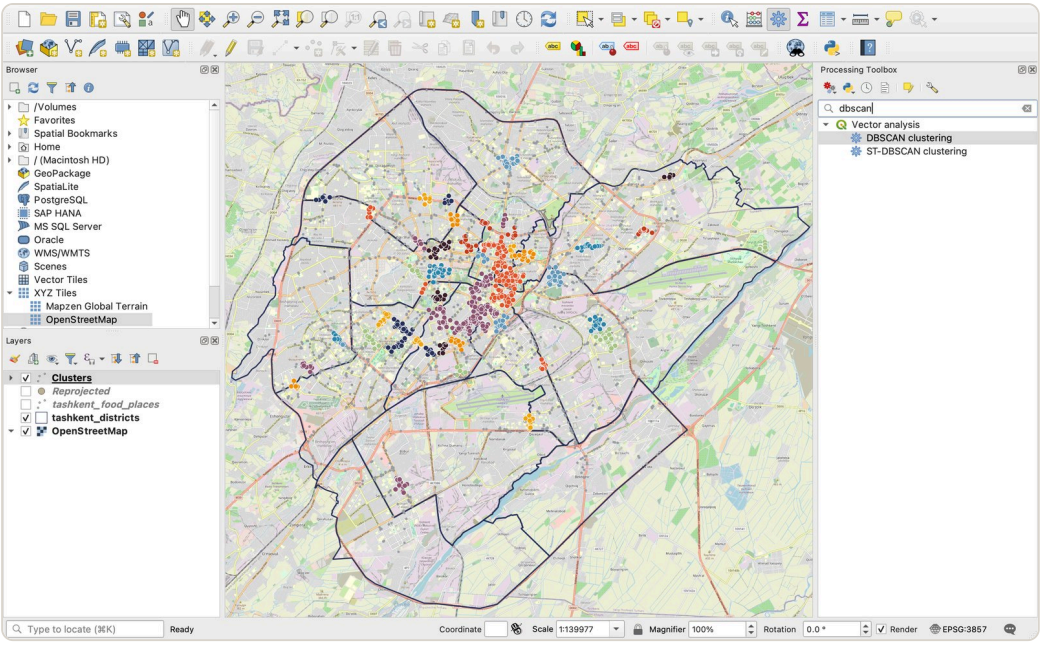


2. Fill in:
    - **Input layer:** `Reprojected`
    - **Minimum cluster size:** `10`
    - **Maximum distance between clustered points:** `300` (meters)
3. Click **Run**, then **Close**.



A new layer **Clusters** appears. Open its attribute table: there are two new columns, `CLUSTER_ID` (which cluster the place belongs to) and `CLUSTER_SIZE` (how many places are in that cluster). Places with `NULL` are noise: not in any cluster.

4. Colour it: double-click **Clusters** → **Symbology** → **Categorized** → **Value:** `CLUSTER_ID` → **Color ramp:** *Random colors* → **Classify** → **OK**. Untick the other point layers to hide them.

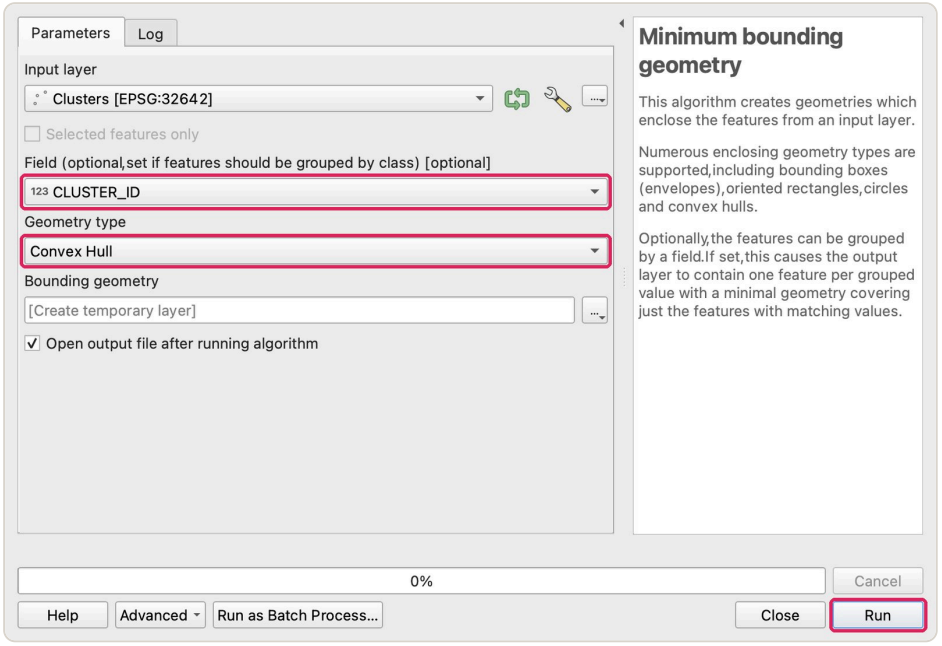


QGIS found **46 clusters**. Together they contain **1,159 of the 2,408 places** (almost half!). The biggest cluster has **243 places**, in the city centre where **Mirobod and Yakkasaroy** districts meet.

> 🧪 **Experiment:** run DBSCAN again with `200` or `500` metres. With 200 m you get smaller, tighter clusters; with 500 m the centre turns into one giant cluster of more than a thousand places. There is no single "right" answer: choose what tells your story best.

🔷 Part 10 · Draw a shape around each cluster

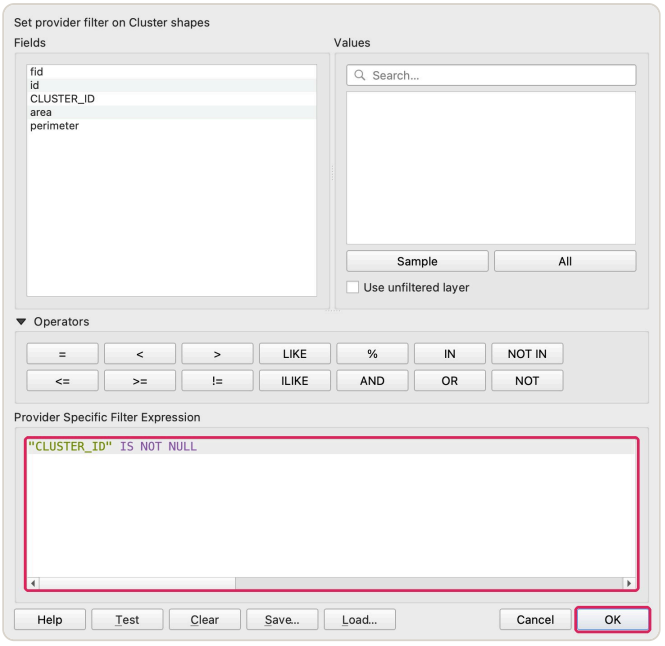
1. In the Processing Toolbox, search `minimum bounding` and double-click **Minimum bounding geometry**.
2. **Input layer:** `Clusters` · **Field:** `CLUSTER_ID` · **Geometry type: Convex Hull**. Click **Run**, then **Close**.



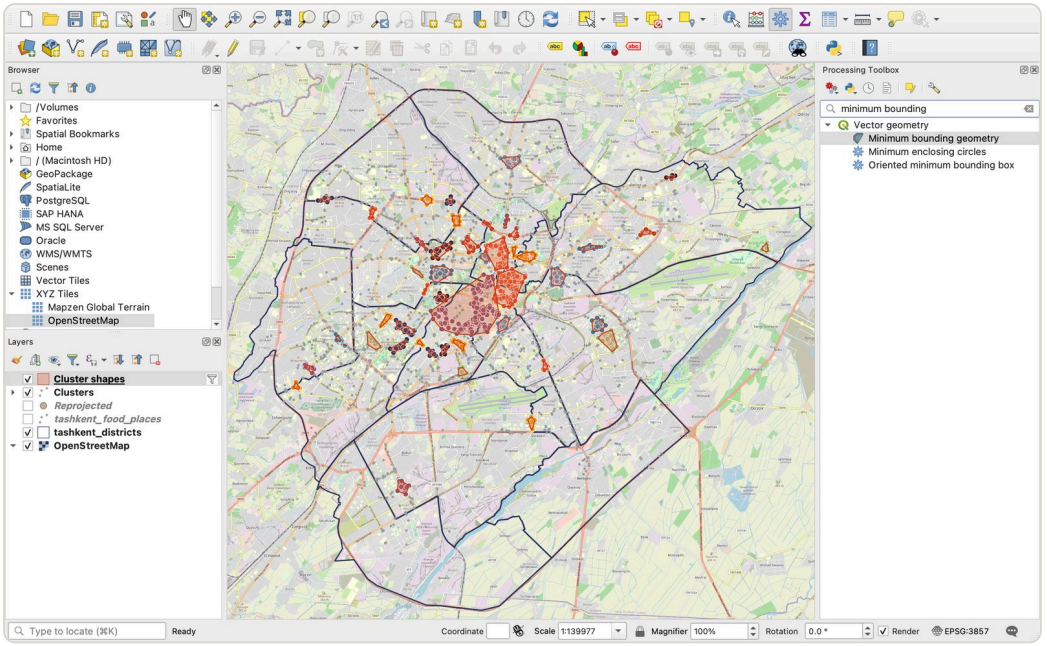
A convex hull is like a rubber band stretched around the points of each cluster. A new layer, **Bounding geometry**, appears (we renamed it *Cluster shapes*: right-click → **Rename Layer**).

3. The noise points (with no cluster) also got one big shape. Remove it: right-click the new layer → **Filter…**, type the expression below and click **OK**:

```
"CLUSTER_ID" IS NOT NULL
```



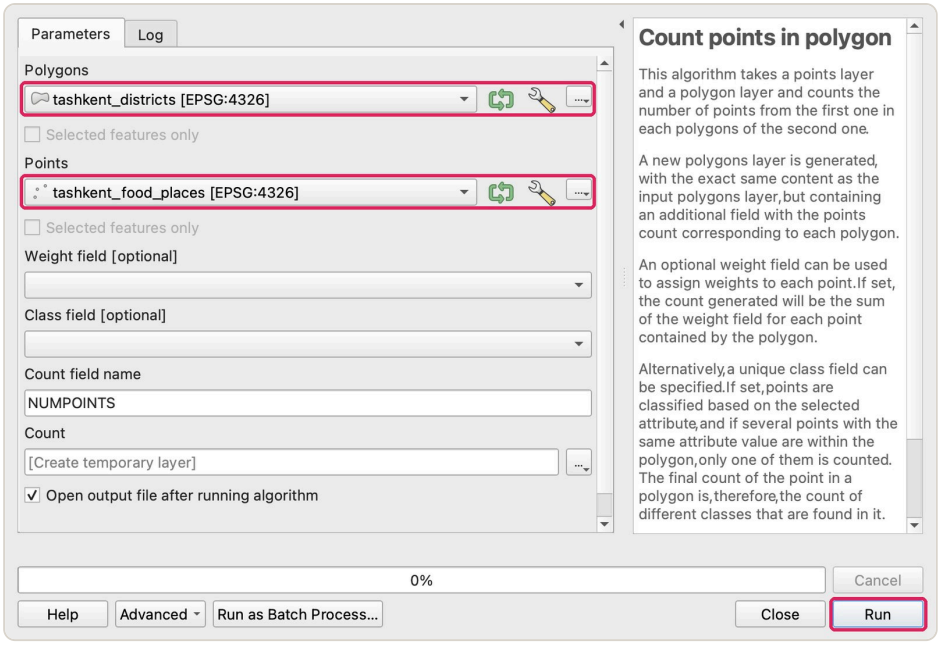
4. Make the shapes see-through: double-click the layer → **Symbology** → set the fill colour's **opacity** to about 30%.



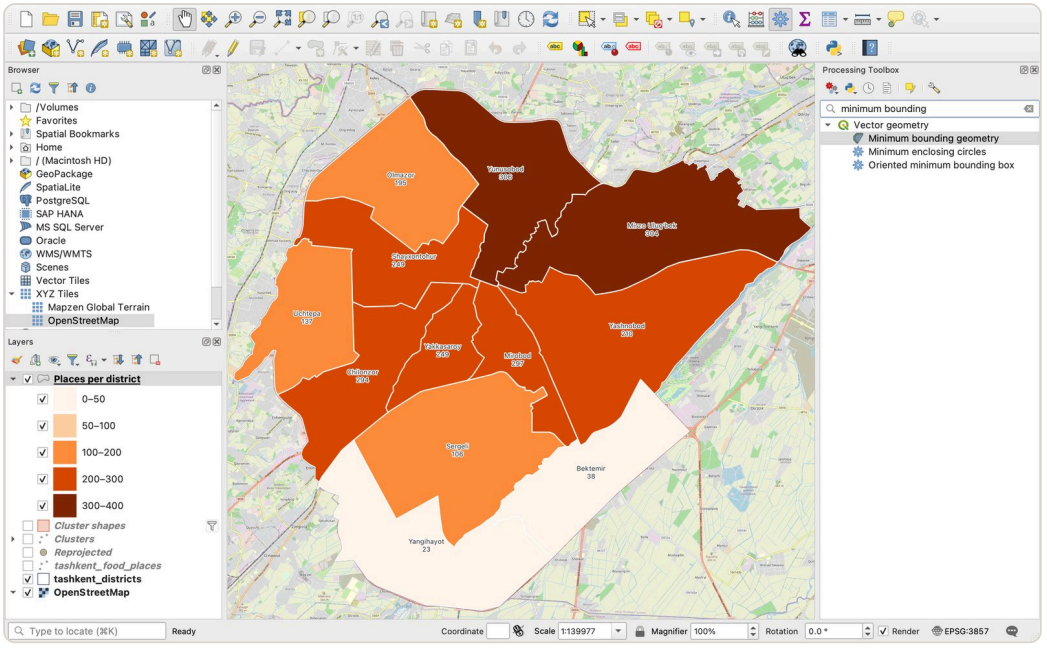
Part 11 · How many places in each district?

1. In the menu, open **Vector** → **Analysis Tools** → **Count Points in Polygon** (or search `count points` in the toolbox).

2. **Polygons:** `tashkent_districts` · **Points:** `tashkent_food_places`. Click **Run**, then **Close**.



The new layer **Count** has a column `NUMPOINTS`. Colour it: **Symbology** → **Graduated** → **Value:** `NUMPOINTS` → pick a colour ramp → **Classify** → **OK**. Add labels: **Labels** → **Single Labels** → **Value:** `district`.

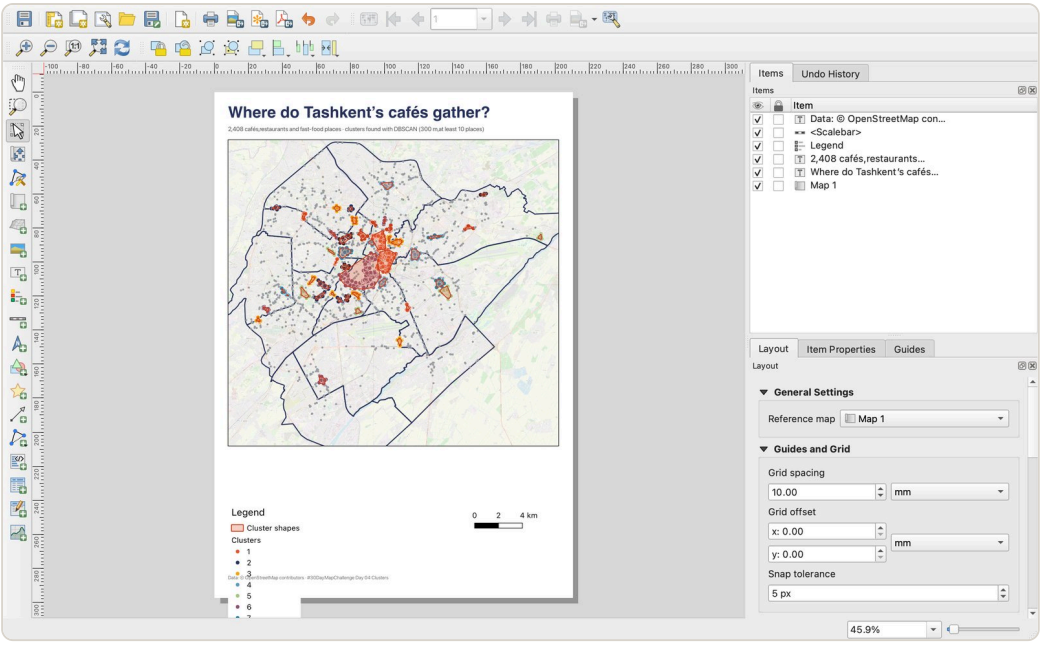


District	Places	Area (km²)	Places per km²
Mirobod	297	16.6	17.9
Yakkasaroy	249	14.0	17.8
Chilonzor	294	30.7	9.6
Shayxontohur	249	27.6	9.0
Yunusobod	306	40.5	7.5
Olmazor	195	33.0	5.9
Mirzo Ulug‘bek	304	55.6	5.5
Uchtepa	137	28.3	4.8
Yashnobod	210	59.1	3.6
Sergeli	106	51.9	2.0
Bektemir	38	31.9	1.2
Yangihayot	23	44.5	0.5

Yunusobod has the most places, but the small central districts **Mirobod** and **Yakkasaroy** are the most crowded: almost 18 places on every km².

🖨️ Part 12 · Make a printable map (Print Layout)

1. **Project** → **New Print Layout**, give it a name (e.g. `Clusters map`) and click **OK**. A new window opens with an empty page.
2. **Add Item** → **Add Map**, then drag a rectangle on the page. Your map appears inside it.
3. **Add Item** → **Add Label** for the title (type it in **Item Properties** on the right and make the font big).
4. **Add Item** → **Add Legend** and **Add Item** → **Add Scale Bar**, and place them under the map.
5. **Layout** → **Export as Image…**, choose a name, and save it as PNG.



💡 In the legend's **Item Properties**, untick **Auto update** to remove layers you don't want (like the 46 cluster numbers). Always add the source: *Data © OpenStreetMap contributors*.

🌈 Make it yours: ideas

- **Other places:** the same steps work for any points. Try `school`, `pharmacy` or `supermarket` from OpenStreetMap (Geofabrik, layer `gis_osm_pois_free_1`) or the fire points from the Day 01 tutorial.
- **Only cafés:** right-click the layer → **Filter…** → `"kind" = 'cafe'`, then run DBSCAN again.
- **Night mode:** remove the OpenStreetMap layer, set the background to dark (**Project** → **Properties** → **General** → **Background color**) and use a glowing colour ramp like *Magma* for the heatmap.
- **Your city:** Samarkand, Bukhara or Namangan? Download the points for Uzbekistan from Geofabrik and zoom to your city.

💾 Share your map

1. Make a folder named after **your map's topic**, e.g. `Coffee clusters of Tashkent`. Put your map inside, plus a short `README.md` with **your name**, the data (OpenStreetMap) and the tool (QGIS).
2. Upload the folder to `2026/Day 04 Clusters/` in this repository. The Quick Start in the main README shows how, with no coding needed.
3. Post it with **#30DayMapChallenge** and credit **© OpenStreetMap contributors**.

🆘 Something went wrong?

What you see	What to do
I can't see the **Processing Toolbox**	**Processing** → **Toolbox** (or **View** → **Panels** → **Processing Toolbox**).
DBSCAN finds **0 clusters** or one huge cluster	You used the layer in degrees. Run DBSCAN on the **Reprojected** layer (Part 8) and check that the distance unit says *meters*.
The districts hide the points	In the Layers panel, drag `tashkent_districts` **below** the point layers, or set its fill to *No Brush* (Part 5).
The OpenStreetMap background stays grey	It needs internet. Check your connection, then zoom a little to reload it.
A tool says the layer is not valid	Make sure you chose the right **Input layer** in the drop-down at the top of the tool.
macOS says QGIS can't be opened	Right-click QGIS in *Applications* → **Open** → **Open**. You only need to do this once.
My layer disappeared	Check the ✓ next to it in the Layers panel, then right-click it → **Zoom to Layer**.

📎 Credits

- Cafés, restaurants and fast food: © OpenStreetMap contributors (ODbL), from the Geofabrik extract of 2 November 2026. See `data/README.md`.
- Tashkent districts: from `Boundaries` in this repository.
- Tool: QGIS, free and open source. DBSCAN: Ester, Kriegel, Sander and Xu (1996).